

Elasticity

ECO101: Principles of Microeconomics

BRAC University

Price Elasticity of Demand

- A measure of the responsiveness of quantity demanded to changes in price.
- Measured by dividing the percentage change in the quantity demanded of a good by the percentage change in its price.

Perfectly Elastic and Perfectly Inelastic Demand

Percentage change in quantity demanded

$$E_d = \frac{\text{Percentage change in quantity demanded}}{\text{Percentage change in price}}$$

- Elastic Demand ($|E_d| > 1$): the numerator is greater than the denominator, the coefficient is greater than 1 and demand is elastic.
- Inelastic Demand ($|E_d| < 1$): the numerator is less than the denominator, the coefficient is less than 1, and demand is inelastic.

Perfectly Elastic and Perfectly Inelastic Demand

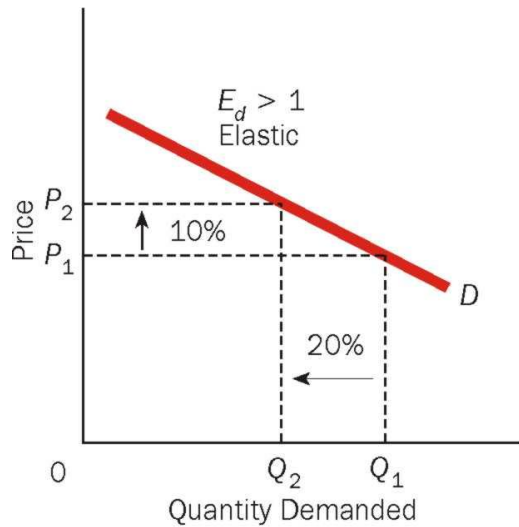
- Unit Elastic Demand ($|E_d| = 1$): If the numerator and denominator are the same, the coefficient is equal to one. The quantity demanded changes proportionally to a change in price.

Elasticity Coefficient	Responsiveness of Quantity Demanded to a Change in Price	Terminology
$E_d > 1$	Quantity demanded changes proportionately more than price changes: $\% \Delta Q_d > \% \Delta P$.	Elastic
$E_d < 1$	Quantity demanded changes proportionately less than price changes: $\% \Delta Q_d < \% \Delta P$.	Inelastic
$E_d = 1$	Quantity demanded changes proportionately to price change: $\% \Delta Q_d = \% \Delta P$.	Unit elastic
$E_d = \infty$	Quantity demanded is extremely responsive to even very small changes in price.	Perfectly elastic
$E_d = 0$	Quantity demanded does not change as price changes.	Perfectly inelastic

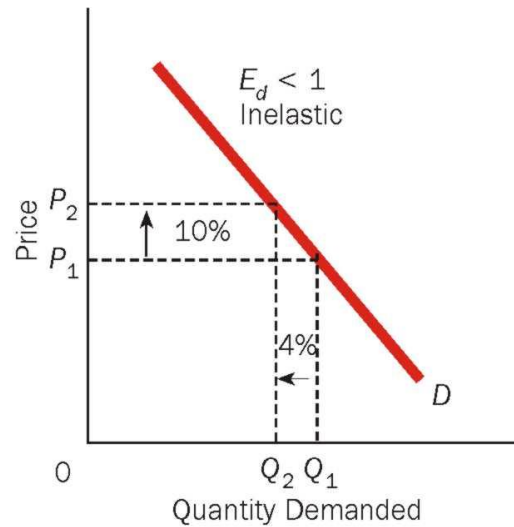
Elastic and Inelastic Demand

- Perfectly Elastic Demand ($E_d = \infty$) If the quantity demanded is extremely responsive to a change in price.
- Perfectly Inelastic Demand ($E_d = 0$) If quantity demanded is completely unresponsive to changes in price, demand is perfectly inelastic. A change in price causes no change in quantity demanded.

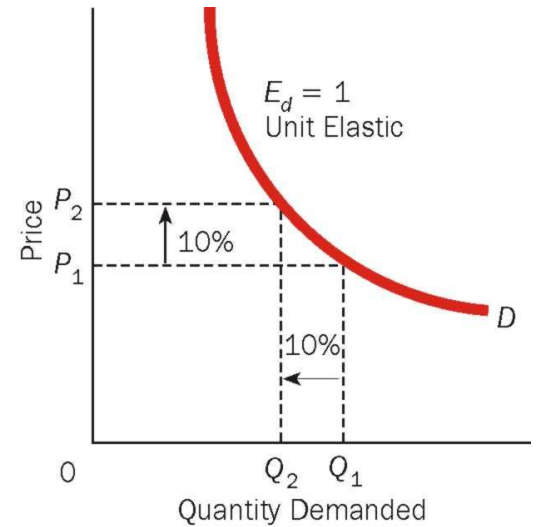
Price Elasticity of Demand



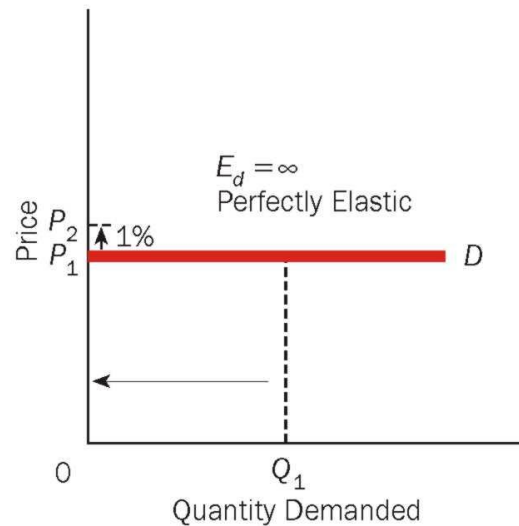
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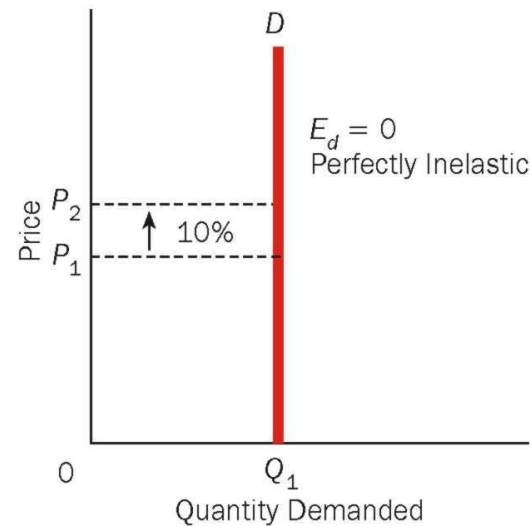
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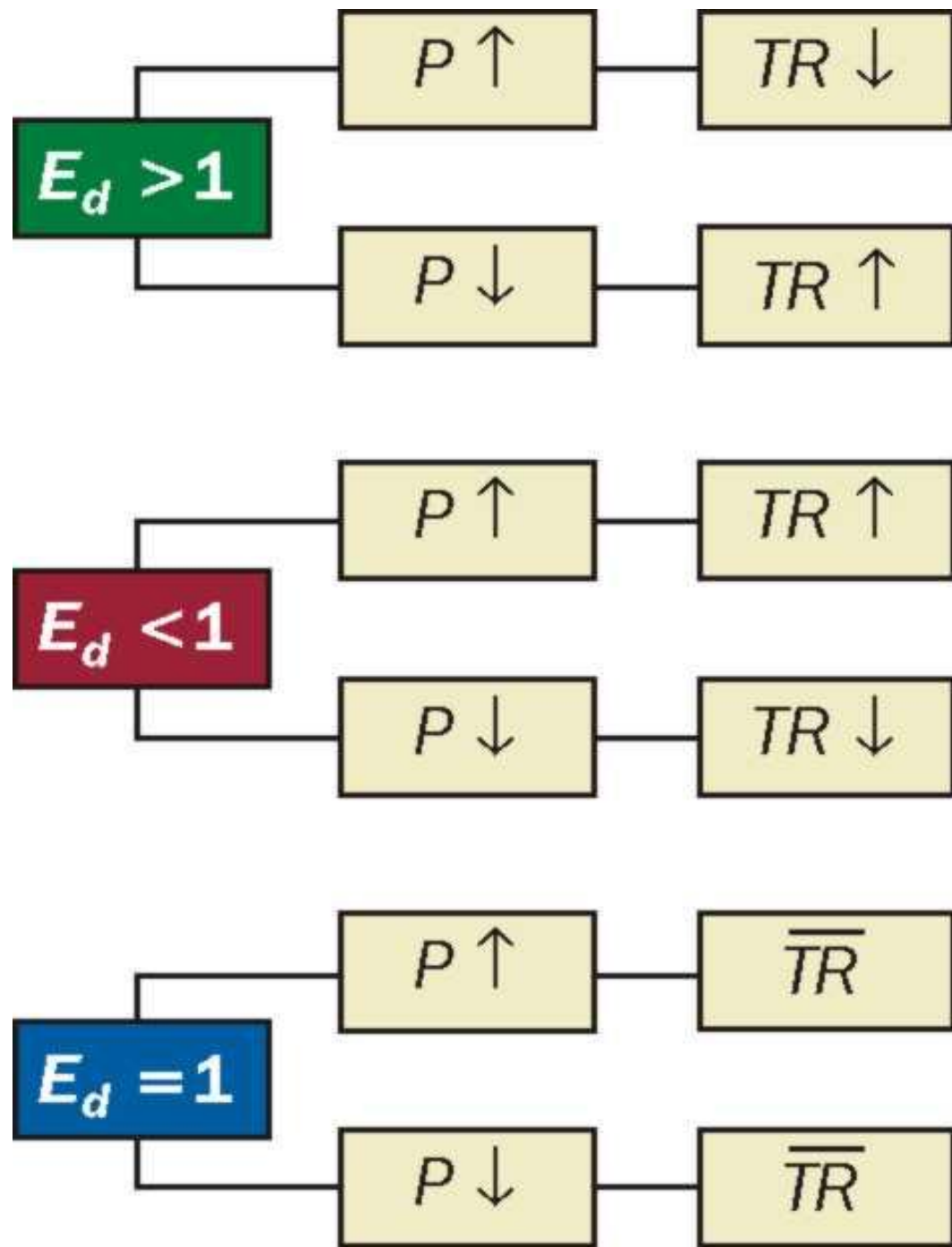


(e)

Price Elasticity of Demand and Total Revenue

- Total Revenue (TR) of a seller equals the price of a good times the quantity of the good sold.
- Total revenue may increase, decrease or remain constant.
- If demand is elastic, a price rise **decreases total revenue**.
- If demand is elastic, a price fall **increases total revenue**.
- If demand is inelastic, a price fall **decreases total revenue**.
- If demand is unit elastic, a price fall will sell more goods while total revenue **remains constant**.

Elasticities, Price Changes and Total Revenue



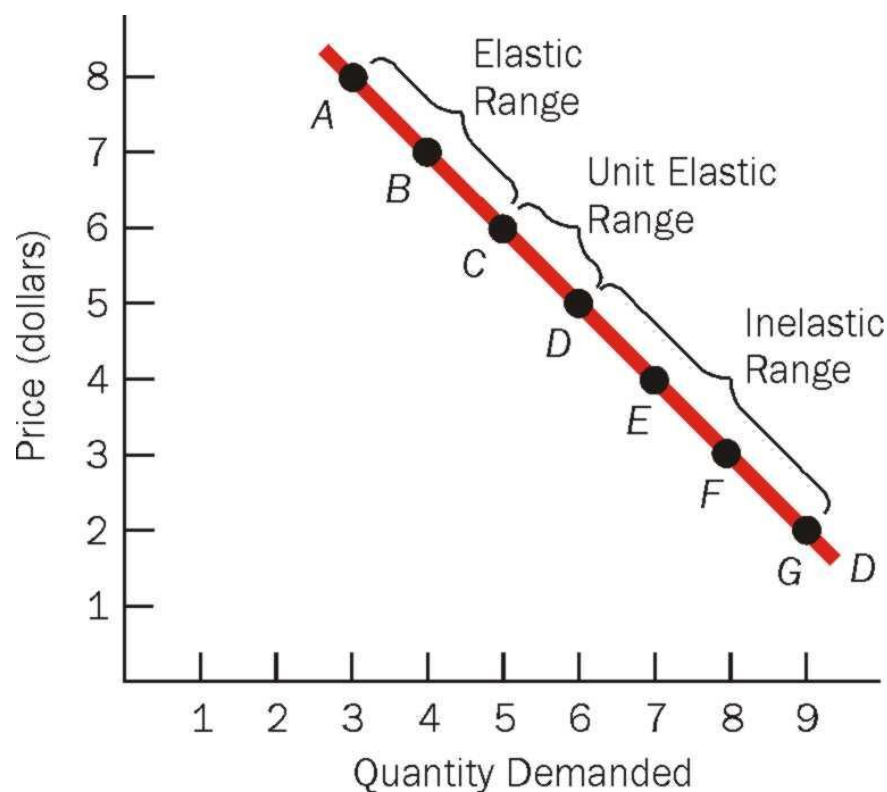
Q & A

- On Tuesday, price and quantity demanded are \$7 and 120 units, respectively. Ten days later, price and quantity are \$6 and 150 units, respectively. What is the price elasticity of demand between the price of \$6 and \$7?
- What does a price elasticity of demand of 0.39 mean?
- Identify what happens to total revenue as a result of each of the following: price rises and demand is elastic; price falls and demand is inelastic; price rises and demand is unit elastic; price rises and demand is inelastic; price falls and demand is elastic.
- Jamal says, “When a seller raises his price, his total revenue rises.” What is Alexi implicitly saying?

Price Elasticity of Demand Along a Straight Line Demand Curve

(1) Point	(2) Price	(3) Quantity Demanded	(4) E_d
A	\$8	3	2.14
B	7	4	1.44
C	6	5	1.00
D	5	6	0.69
E	4	7	0.47
F	3	8	0.29
G	2	9	

(a)



(b)

Determinants of Price Elasticity on Demand

- **Number of Substitutes:** The more substitutes for a good, the higher the price elasticity of demand; the fewer substitutes for a good, the lower the price elasticity of demand. The more broadly defined the good, the fewer the substitutes; the more narrowly defined the good, the greater the substitutes.
- **Necessities Versus Luxuries:** The more that a good is considered a luxury rather than a necessity, the higher the price elasticity of demand.

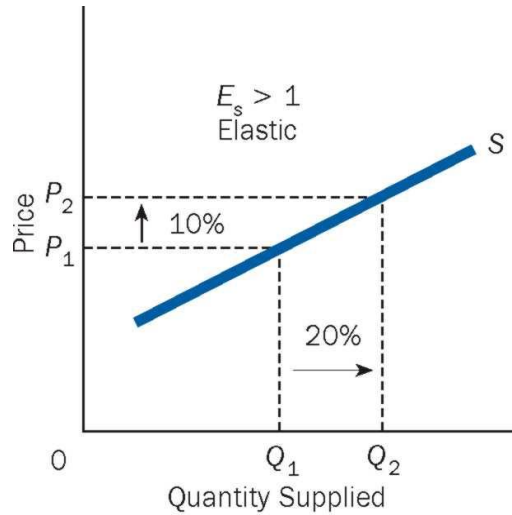
Q & A

- If there are 7 substitutes for good X and demand is inelastic, does it follow that if there are 9 substitutes for good X demand will be elastic? Explain your answer.
- Price elasticity of demand is predicted to be higher for which good of the following combinations of goods: Compaq computers or computers; Heinz ketchup or ketchup; Perrier water or water? Explain your answers.

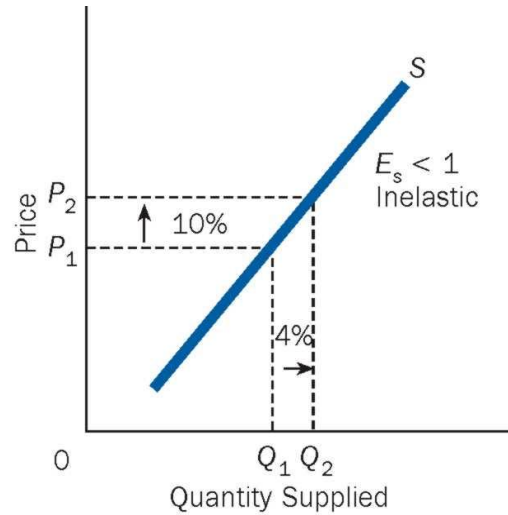
Price Elasticity of Supply

- Measures the responsiveness of quantity supplied to changes in price.
- Defined as the percentage change in quantity supplied of a good divided by the percentage change in the price of the good.
- Supply can be classified as elastic, inelastic, unit elastic, perfectly elastic, or perfectly inelastic.

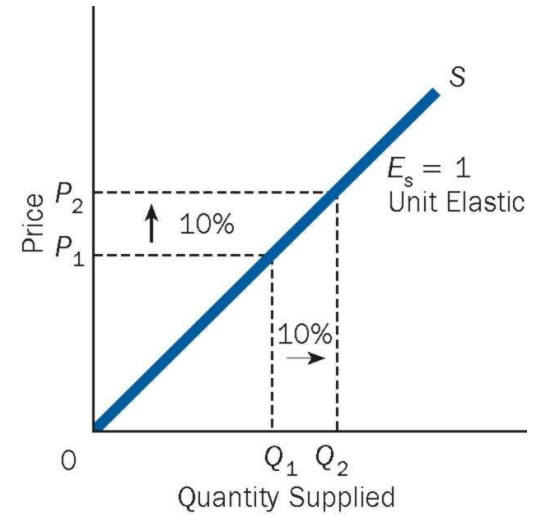
Price Elasticity of Supply



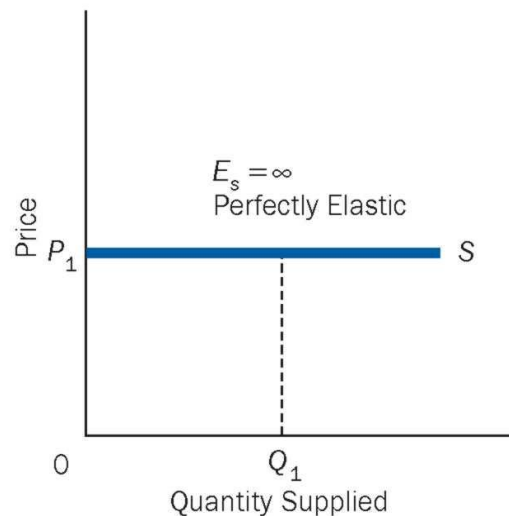
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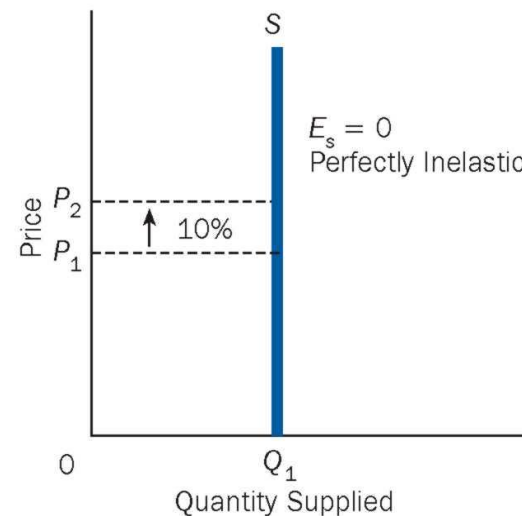
(b)



(c)



(d)



(e)

Price Elasticity of Supply and Time

- The longer the period of adjustment to a change in price, the higher the price elasticity of supply.
- Additional production takes time.
- Reducing production takes time.

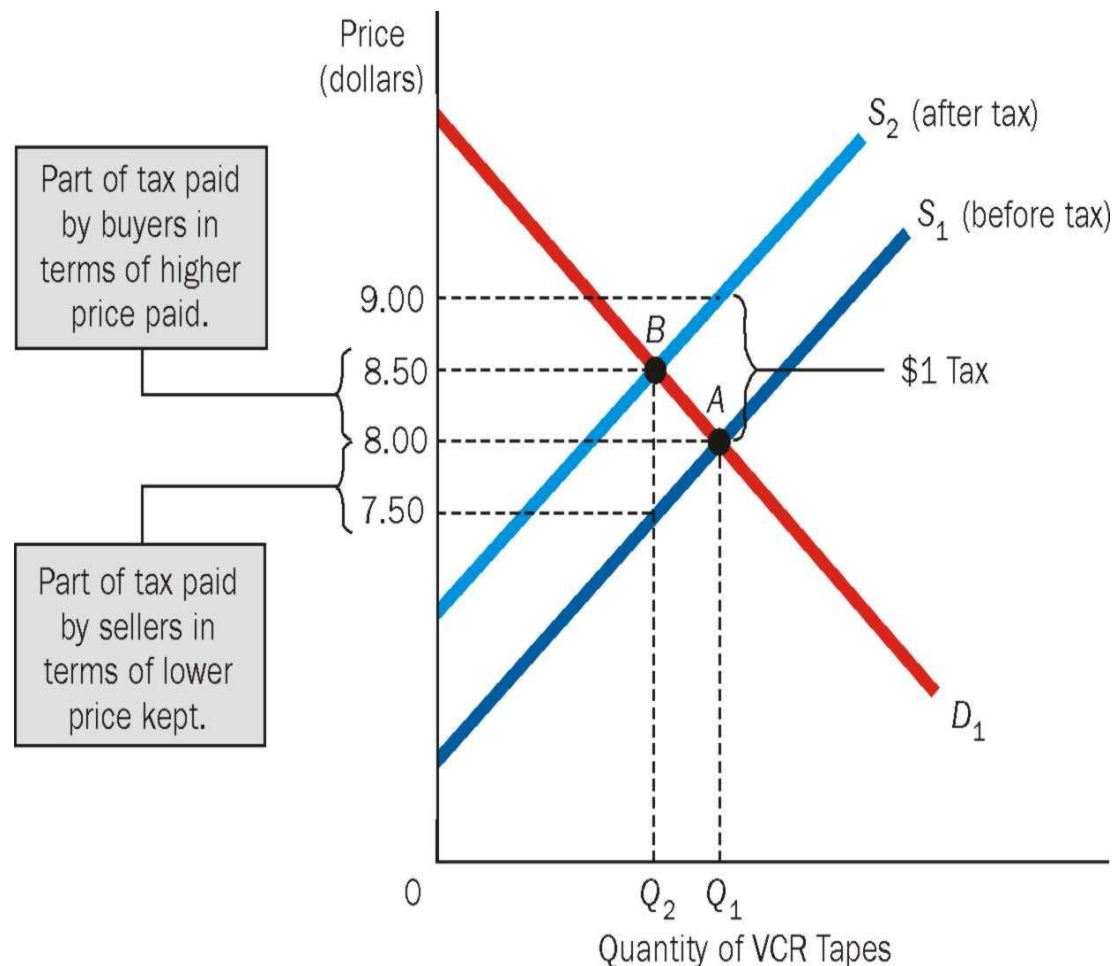


Summary of the Four Elasticity Concepts

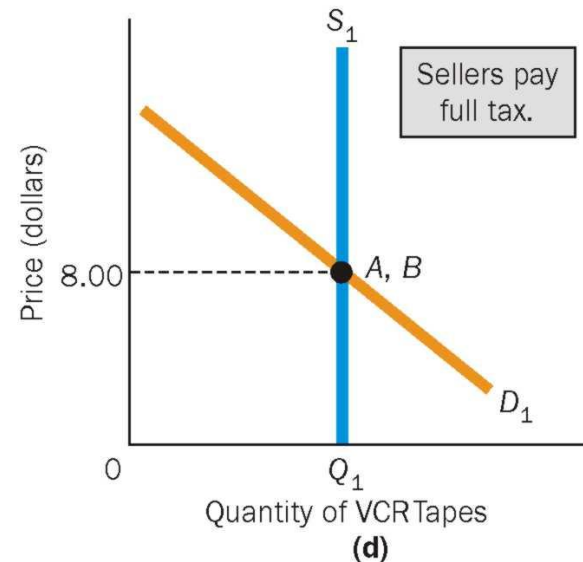
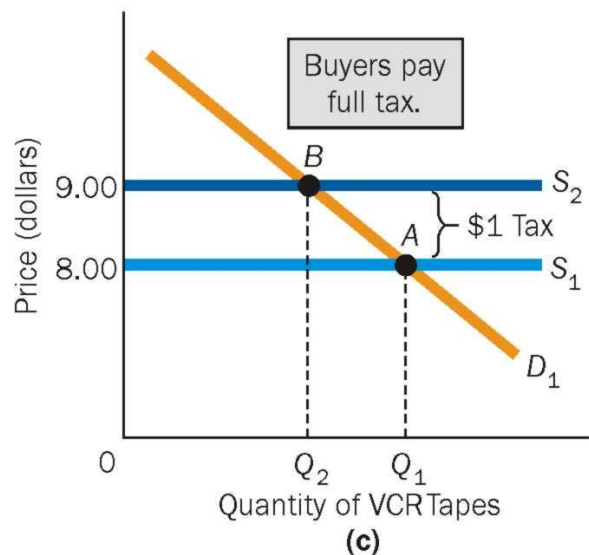
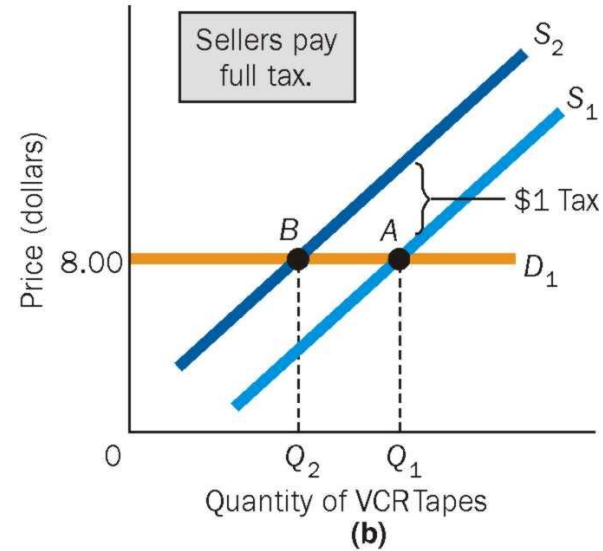
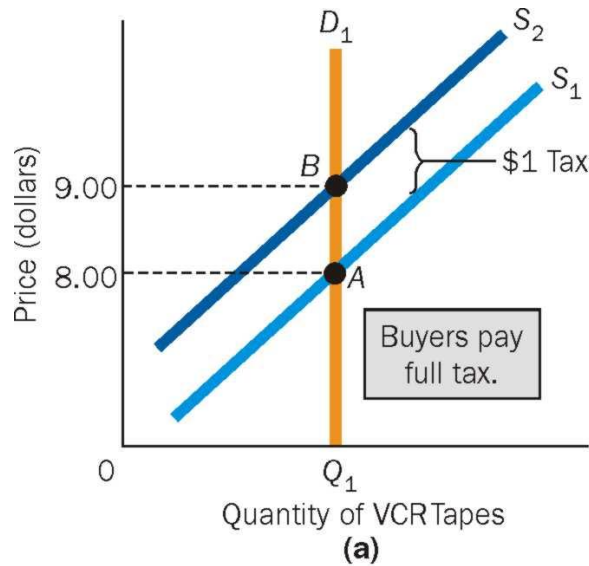
Type	Definition	Possibilities	Terminology
Price elasticity of demand	$\frac{\text{Percentage change in quantity demanded}}{\text{Percentage change in price}}$	$E_d > 1$ $E_d < 1$ $E_d = 1$ $E_d = \infty$ $E_d = 0$	Elastic Inelastic Unit elastic Perfectly elastic Perfectly inelastic
Cross elasticity of demand	$\frac{\text{Percentage change in quantity demanded of one good}}{\text{Percentage change in price of another good}}$	$E_c < 0$ $E_c > 0$	Complements Substitutes
Income elasticity of demand	$\frac{\text{Percentage change in quantity demanded}}{\text{Percentage change in income}}$	$E_y > 0$ $E_y < 0$ $E_y > 1$ $E_y < 1$ $E_y = 1$	Normal good Inferior good Income elastic Income inelastic Income unit elastic
Price elasticity of supply	$\frac{\text{Percentage change in quantity supplied}}{\text{Percentage change in price}}$	$E_s > 1$ $E_s < 1$ $E_s = 1$ $E_s = \infty$ $E_s = 0$	Elastic Inelastic Unit elastic Perfectly elastic Perfectly inelastic

Who Pays the Tax?

- A tax placed on the sellers of VCR tapes shifts the supply curve from S_1 to S_2 and raises the equilibrium price from \$8 to \$8.50. Part of the tax is paid by buyers through a higher price paid, and part of the tax is paid by sellers through a lower price kept.
- Tax revenues are maximized by placing the tax on the seller who faces the more inelastic demand curve.



Different Elasticities and Who Pays the Tax



Q & A

- What does an income elasticity of demand of 1.33 mean?
- If supply is perfectly inelastic, what does this signify?
- Why will government raise more tax revenue if it applies a tax to a good with inelastic demand than if it applies the tax to a good with elastic demand?
- Under what condition would a per-unit tax placed on the sellers of computers be fully paid by the buyers of computers?

Cross Elasticity of Demand

- Measures the responsiveness in the quantity demanded of one good to changes in the price of another good.
- Defined as the percentage change in the quantity demanded of one good divided by the percentage change in the price of another good.
- This concept is often used to determine whether two goods are substitutes or complements and the degree to which one good is a complement to or substitute for another.

Income Elasticity of Demand

- Measures the responsiveness of quantity demanded to changes in income.
- Define as the percentage change in quantity demanded of a good divided by the percentage change in income.
- Income elasticity of demand is positive ($E_y > 0$) for a normal good.
- The demand for an inferior good decreases as income increases.

Income Elasticity of Demand



- If $E_y > 1$, demand is considered to be income elastic.
- If $E_y < 1$, demand is considered to be income inelastic.
- If $E_y = 1$, demand is considered to be unit elastic.